

Week 4: Order Management System

Scenario:

The e-commerce company needs a system to manage customer orders. Customers can purchase multiple products, and each order should maintain details such as order date, purchased products, quantity, price, and total order amount.

1. Design Orders and Order_Details Tables

Orders Entity (Stores overall order information.)

Attribute	Description
Order_ID	Unique order identifier
Customer_ID	Customer reference
Order_Date	Date of order placement
Total_Amount	Total bill amount
Order_Status	Pending/Shipped/Delivered

```
SQL> CREATE TABLE OrdersEntity(OrderID VARCHAR2(7),CustomerID VARCHAR2(7),
  2  OrderDate DATE,TotalAmount number(9,2),OrderStatus VARCHAR2(15), PRIMARY KEY(OrderID),
  3  FOREIGN KEY(CustomerID) REFERENCES Customer(Customer_ID));

Table created.
```

Orders table has been created with Foreign Key CustomerID and Primary Key as Order ID.

Order_Details Entity

Stores individual products included in an order

. Attribute	Description
Order_Detail_ID	Unique detail identifier
Order_ID	Order reference
Product_ID	Product reference
Quantity	Number of products purchased
Price	Product price

```
SQL> CREATE TABLE OrderDetail(Order_Detail_ID VARCHAR2(7),
2 OrderID VARCHAR2(7), ProductID VARCHAR2(7),Quantity number,
3 Price number(9,2), PRIMARY KEY(Order_Detail_ID),
4 FOREIGN KEY(OrderID) REFERENCES OrdersEntity(OrderID), FOREIGN KEY(ProductID) REFERENCES Product(Product_ID));
Table created.
```

2. Establish Relationships

Customer and Orders

- One customer can place multiple orders.
- Each order belongs to one customer.

Relationship:

Customer (1) ----- (Many) Orders

Orders and Products

- One order can contain multiple products.
- One product can appear in multiple orders.

Relationship:

Orders (Many) ----- (Many) Product

This relationship is implemented using:

Order_Details Table

Apply Constraints:

- Primary Key
- Foreign Key
- NOT NULL
- CHECK

Apply Constraints:

```
SQL> ALTER TABLE OrderDetail
2 ADD CONSTRAINT CHK_Quantity CHECK (Quantity >= 0);
Table altered.
```

Likewise apply constraints

3. Manage Customer Product Orders

Tasks:

1. Create a new order for a customer.
2. Add multiple products to an order.
3. Calculate total order amount.
4. Update order status.
5. View complete order details.

Orders Table

```
SQL> INSERT INTO ORDERS (ORDERID, CUSTOMERID, ORDERDATE, TOTALAMOUNT, ORDERSTATUS)
  2  VALUES ('ORD0001', 'CUST000001', DATE '2026-01-10', 55850.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO ORDERS (ORDERID, CUSTOMERID, ORDERDATE, TOTALAMOUNT, ORDERSTATUS)
  2  VALUES ('ORD0002', 'CUST000002', DATE '2026-01-25', 3350.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO ORDERS (ORDERID, CUSTOMERID, ORDERDATE, TOTALAMOUNT, ORDERSTATUS)
  2  VALUES ('ORD0003', 'CUST000003', DATE '2026-02-05', 28000.00, 'Shipped');

1 row created.

SQL>
SQL> INSERT INTO ORDERS (ORDERID, CUSTOMERID, ORDERDATE, TOTALAMOUNT, ORDERSTATUS)
  2  VALUES ('ORD0004', 'CUST000004', DATE '2026-02-18', 8300.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO ORDERS (ORDERID, CUSTOMERID, ORDERDATE, TOTALAMOUNT, ORDERSTATUS)
  2  VALUES ('ORD0005', 'CUST000005', DATE '2026-03-02', 22000.00, 'Processing');

1 row created.

SQL>
SQL> INSERT INTO ORDERS (ORDERID, CUSTOMERID, ORDERDATE, TOTALAMOUNT, ORDERSTATUS)
  2  VALUES ('ORD0006', 'CUST000006', DATE '2026-03-15', 7000.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO ORDERS (ORDERID, CUSTOMERID, ORDERDATE, TOTALAMOUNT, ORDERSTATUS)
  2  VALUES ('ORD0007', 'CUST000007', DATE '2026-04-01', 6500.00, 'Shipped');

1 row created.
```

```
SQL> select * from orders;
```

ORDERID	CUSTOMERID	ORDERDATE	TOTALAMOUNT	ORDERSTATUS
ORD0001	CUST000001	10-JAN-26	55850	Delivered
ORD0002	CUST000002	25-JAN-26	3350	Delivered
ORD0003	CUST000003	05-FEB-26	28000	Shipped
ORD0004	CUST000004	18-FEB-26	8300	Delivered
ORD0005	CUST000005	02-MAR-26	22000	Processing
ORD0006	CUST000006	15-MAR-26	7000	Delivered
ORD0007	CUST000007	01-APR-26	6500	Shipped
ORD0008	CUST000008	20-APR-26	5700	Delivered
ORD0009	CUST000009	05-MAY-26	18500	Processing
ORD0010	CUST000010	18-MAY-26	5200	Delivered
ORD0011	CUST000011	03-JUN-26	7000	Shipped
ORD0012	CUST000012	20-JUN-26	2450	Delivered
ORD0013	CUST000013	08-JUL-26	3500	Cancelled
ORD0014	CUST000014	15-AUG-26	28000	Delivered
ORD0015	CUST000015	25-SEP-26	8350	Processing

```
15 rows selected.
```

OrderDetails Table

```
SQL> INSERT INTO ORDERDETAIL
  2 (ORDER_DETAIL_ID, ORDERID, PRODUCTID, QUANTITY, PRICE)
  3 VALUES ('OD00011', 'ORD0007', 'P000008', 1, 6500.00);

1 row created.

SQL>
SQL> INSERT INTO ORDERDETAIL
  2 (ORDER_DETAIL_ID, ORDERID, PRODUCTID, QUANTITY, PRICE)
  3 VALUES ('OD00012', 'ORD0008', 'P000012', 1, 5200.00);

1 row created.

SQL>
SQL> INSERT INTO ORDERDETAIL
  2 (ORDER_DETAIL_ID, ORDERID, PRODUCTID, QUANTITY, PRICE)
  3 VALUES ('OD00013', 'ORD0008', 'P000006', 1, 450.00);

1 row created.

SQL>
SQL> INSERT INTO ORDERDETAIL
  2 (ORDER_DETAIL_ID, ORDERID, PRODUCTID, QUANTITY, PRICE)
  3 VALUES ('OD00014', 'ORD0009', 'P000013', 1, 18500.00);

1 row created.

SQL>
SQL> INSERT INTO ORDERDETAIL
  2 (ORDER_DETAIL_ID, ORDERID, PRODUCTID, QUANTITY, PRICE)
  3 VALUES ('OD00015', 'ORD0010', 'P000012', 1, 5200.00);

1 row created.
```

```
SQL> SELECT * FROM OrderDetail;
```

ORDER_D	ORDERID	PRODUCT	QUANTITY	PRICE
OD00001	ORD0001	P000001	1	55000
OD00002	ORD0001	P000002	1	850
OD00003	ORD0002	P000004	1	3200
OD00004	ORD0002	P000006	1	450
OD00005	ORD0003	P000005	1	28000
OD00006	ORD0004	P000008	1	6500
OD00007	ORD0004	P000006	4	450
OD00008	ORD0005	P000010	1	22000
OD00009	ORD0006	P000009	1	4200
OD00010	ORD0006	P000015	1	3500
OD00011	ORD0007	P000008	1	6500

ORDER_D	ORDERID	PRODUCT	QUANTITY	PRICE
OD00012	ORD0008	P000012	1	5200
OD00013	ORD0008	P000006	1	450
OD00014	ORD0009	P000013	1	18500
OD00015	ORD0010	P000012	1	5200
OD00016	ORD0011	P000009	1	4200
OD00017	ORD0011	P000015	1	3500
OD00018	ORD0012	P000014	1	1500
OD00019	ORD0012	P000003	1	2500
OD00020	ORD0015	P000015	1	3500

20 rows selected.

Update Operation

Modify order information.

Examples:

- Change order status from Pending to Delivered.

```
SQL> UPDATE Orders SET OrderStatus = 'Delivered' WHERE OrderID = 'ORD0003';
```

1 row updated.

```
SQL> SELECT * FROM Orders;
```

ORDERID	CUSTOMERID	ORDERDATE	TOTALAMOUNT	ORDERSTATUS
ORD0001	CUST000001	10-JAN-26	55850	Delivered
ORD0002	CUST000002	25-JAN-26	3350	Delivered
ORD0003	CUST000003	05-FEB-26	28000	Delivered
ORD0004	CUST000004	18-FEB-26	8300	Delivered
ORD0005	CUST000005	02-MAR-26	22000	Processing
ORD0006	CUST000006	15-MAR-26	7000	Delivered
ORD0007	CUST000007	01-APR-26	6500	Shipped
ORD0008	CUST000008	20-APR-26	5700	Delivered
ORD0009	CUST000009	05-MAY-26	18500	Processing
ORD0010	CUST000010	18-MAY-26	5200	Delivered
ORD0011	CUST000011	03-JUN-26	7000	Shipped
ORD0012	CUST000012	20-JUN-26	2450	Delivered
ORD0013	CUST000013	08-JUL-26	3500	Cancelled
ORD0014	CUST000014	15-AUG-26	28000	Delivered
ORD0015	CUST000015	25-SEP-26	8350	Processing

```
15 rows selected.
```

- Update product quantity.

```
SQL> UPDATE OrderDetail SET Quantity = 6 WHERE OrderID = 'ORD0003';
```

```
1 row updated.
```

```
SQL> SELECT * FROM OrderDetail;
```

ORDER_D	ORDERID	PRODUCT	QUANTITY	PRICE
OD00001	ORD0001	P000001	1	55000
OD00002	ORD0001	P000002	1	850
OD00003	ORD0002	P000004	1	3200
OD00004	ORD0002	P000006	1	450
OD00005	ORD0003	P000005	6	28000
OD00006	ORD0004	P000008	1	6500
OD00007	ORD0004	P000006	4	450
OD00008	ORD0005	P000010	1	22000
OD00009	ORD0006	P000009	1	4200
OD00010	ORD0006	P000015	1	3500
OD00011	ORD0007	P000008	1	6500

ORDER_D	ORDERID	PRODUCT	QUANTITY	PRICE
OD00012	ORD0008	P000012	1	5200
OD00013	ORD0008	P000006	1	450
OD00014	ORD0009	P000013	1	18500
OD00015	ORD0010	P000012	1	5200
OD00016	ORD0011	P000009	1	4200
OD00017	ORD0011	P000015	1	3500
OD00018	ORD0012	P000014	1	1500
OD00019	ORD0012	P000003	1	2500
OD00020	ORD0015	P000015	1	3500

```
20 rows selected.
```

Delete Operation

Remove cancelled orders.

Example:

"Delete orders cancelled by customers."

```
SQL> DELETE FROM Orders WHERE OrderStatus = 'Cancelled';
```

```
1 row deleted.
```

```
SQL> SELECT * FROM Orders;
```

ORDERID	CUSTOMERID	ORDERDATE	TOTALAMOUNT	ORDERSTATUS
ORD0001	CUST000001	10-JAN-26	55850	Delivered
ORD0002	CUST000002	25-JAN-26	3350	Delivered
ORD0003	CUST000003	05-FEB-26	28000	Delivered
ORD0004	CUST000004	18-FEB-26	8300	Delivered
ORD0005	CUST000005	02-MAR-26	22000	Processing
ORD0006	CUST000006	15-MAR-26	7000	Delivered
ORD0007	CUST000007	01-APR-26	6500	Shipped
ORD0008	CUST000008	20-APR-26	5700	Delivered
ORD0009	CUST000009	05-MAY-26	18500	Processing
ORD0010	CUST000010	18-MAY-26	5200	Delivered
ORD0011	CUST000011	03-JUN-26	7000	Shipped

ORDERID	CUSTOMERID	ORDERDATE	TOTALAMOUNT	ORDERSTATUS
ORD0012	CUST000012	20-JUN-26	2450	Delivered
ORD0014	CUST000014	15-AUG-26	28000	Delivered
ORD0015	CUST000015	25-SEP-26	8350	Processing

```
14 rows selected.
```

5. Generate Customer Order History Reports

Write SQL queries to generate reports.

Report 1: Customer Order History

Display:

- Customer Name
- Order ID
- Order Date
- Total Amount
- Status

```
SQL> SELECT CONCAT(First_Name,Last_Name) AS CustomerName,O.OrderID,O.OrderDate,O.TotalAmount,O.OrderStatus
2 FROM Customer C JOIN Orders O ON
3 C.Customer_ID = O.CustomerID;
```

CUSTOMERNAME	ORDERID	ORDERDATE	TOTALAMOUNT
ArjunKumar Delivered	ORD0001	10-JAN-26	55850
PriyaSharma Delivered	ORD0002	25-JAN-26	3350
RahulMenon Delivered	ORD0003	05-FEB-26	28000

CUSTOMERNAME	ORDERID	ORDERDATE	TOTALAMOUNT
AnanyaIyer Delivered	ORD0004	18-FEB-26	8300

VikramReddy Processing	ORD0005	02-MAR-26	22000
MeeraKrishnan Delivered	ORD0006	15-MAR-26	7000

CUSTOMERNAME	ORDERID	ORDERDATE	TOTALAMOUNT
SureshNair Shipped	ORD0007	01-APR-26	6500

KavyaRajan Delivered	ORD0008	20-APR-26	5700
KarthikSubramanian Processing	ORD0009	05-MAY-26	18500

CUSTOMERNAME	ORDERID	ORDERDATE	TOTALAMOUNT
DivyaPillai Delivered	ORD0010	18-MAY-26	5200

AdityaVerma Shipped	ORD0011	03-JUN-26	7000
SnehaPatel Delivered	ORD0012	20-JUN-26	2450

CUSTOMERNAME	ORDERID	ORDERDATE	TOTALAMOUNT
ORDERSTATUS			
LakshmiVenkatesan Delivered	ORD0014	15-AUG-26	28000
NithinMurugan Processing	ORD0015	25-SEP-26	8350

Report 2: Product-wise Order Report

Find:

- Number of times each product was ordered.
- Total quantity sold.

```
SQL> SELECT p.Product_ID,
2 SUM(od.Quantity) AS Quantity,
3 COUNT(od.ProductID) AS OrderTimes
4 FROM Product p JOIN
5 OrderDetail od ON
6 p.Product_ID = od.ProductID
7 GROUP BY od.ProductID;
```

PRODUCT	QUANTITY	ORDERTIMES
P000001	1	1
P000002	1	1
P000004	1	1
P000006	6	3
P000005	6	1
P000008	2	2
P000010	1	1
P000009	2	2
P000015	3	3
P000012	2	2
P000013	1	1

PRODUCT	QUANTITY	ORDERTIMES
P000014	1	1
P000003	1	1

13 rows selected.

Report 3: Customer Purchase Analysis

Customer Order Report

```
SQL> SELECT
  2  c.Customer_ID,
  3  CONCAT(c.First_Name,c.Last_Name) AS CustomerName,
  4  o.OrderID,
  5  o.OrderDate,
  6  od. ProductID,
  7  od.Quantity,
  8  od.Price,
  9  od.Price * od.Quantity AS Subtotal
 10 FROM Customer c
 11 JOIN Orders o
 12 ON c.Customer_ID = o.CustomerID
 13 JOIN OrderDetail od
 14 ON o.OrderID = od.OrderID
 15 ORDER BY c.Customer_ID, o.OrderDate;
```

CUSTOMER_I	CUSTOMERNAME	ORDERID	ORDERDATE	PRODUCT
QUANTITY	PRICE	SUBTOTAL		
CUST000001	ArjunKumar	ORD0001	10-JAN-26	P000001
1	55000			
				55000
CUST000001	ArjunKumar	ORD0001	10-JAN-26	P000002
1	850			
				850
CUST000002	PriyaSharma	ORD0002	25-JAN-26	P000004
1	3200			
				3200

CUSTOMER_I	CUSTOMERNAME	ORDERID	ORDERDATE	PRODUCT
QUANTITY	PRICE	SUBTOTAL		
CUST000002	PriyaSharma	ORD0002	25-JAN-26	P000006
1	450			
				450
CUST000003	RahulMenon	ORD0003	05-FEB-26	P000005
6	28000			
				168000
CUST000004	AnanyaIyer	ORD0004	18-FEB-26	P000008
1	6500			
				6500

CUSTOMER_I	CUSTOMERNAME	ORDERID	ORDERDATE	PRODUCT
QUANTITY	PRICE	SUBTOTAL		
CUST000004	AnanyaIyer	ORD0004	18-FEB-26	P000006
4	450	1800		

CUST000005	VikramReddy	ORD0005	02-MAR-26	P000010
1	22000	22000		

CUST000006	MeeraKrishnan	ORD0006	15-MAR-26	P000009
1	4200	4200		

CUSTOMER_I	CUSTOMERNAME	ORDERID	ORDERDATE	PRODUCT
QUANTITY	PRICE	SUBTOTAL		
CUST000006	MeeraKrishnan	ORD0006	15-MAR-26	P000015
1	3500	3500		

CUST000007	SureshNair	ORD0007	01-APR-26	P000008
1	6500	6500		

CUST000008	KavyaRajan	ORD0008	20-APR-26	P000012
1	5200	5200		

CUSTOMER_I	CUSTOMERNAME	ORDERID	ORDERDATE	PRODUCT
QUANTITY	PRICE	SUBTOTAL		
CUST000008	KavyaRajan	ORD0008	20-APR-26	P000006
1	450	450		

CUST000009	KarthikSubramanian	ORD0009	05-MAY-26	P000013
1	18500	18500		

CUST000010	DivyaPillai	ORD0010	18-MAY-26	P000012
1	5200	5200		

CUSTOMER_I	CUSTOMERNAME	ORDERID	ORDERDATE	PRODUCT
QUANTITY	PRICE	SUBTOTAL		
CUST000011	AdityaVerma	ORD0011	03-JUN-26	P000009
1	4200	4200		
CUST000011	AdityaVerma	ORD0011	03-JUN-26	P000015
1	3500	3500		
CUST000012	SnehaPatel	ORD0012	20-JUN-26	P000014
1	1500	1500		
CUSTOMER_I	CUSTOMERNAME	ORDERID	ORDERDATE	PRODUCT
QUANTITY	PRICE	SUBTOTAL		
CUST000012	SnehaPatel	ORD0012	20-JUN-26	P000003
1	2500	2500		
CUST000015	NithinMurugan	ORD0015	25-SEP-26	P000015
1	3500	3500		

20 rows selected.

```

SQL> SELECT
  2  c.Customer_ID,
  3  CONCAT(c.First_Name,c.Last_Name) AS Customer_Name,
  4  COUNT(o.OrderID) AS Total_Orders,
  5  SUM(o.TotalAmount) AS Total_Amount
  6  FROM Customer c
  7  JOIN Orders o
  8  ON c.Customer_ID = o.CustomerID
  9  GROUP BY c.Customer_ID, CONCAT(c.First_Name,c.Last_Name)
 10  ORDER BY c.Customer_ID;

```

CUSTOMER_I	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_AMOUNT
CUST000001	ArjunKumar	1	55850
CUST000002	PriyaSharma	1	3350
CUST000003	RahulMenon	1	28000
CUST000004	AnanyaIyer	1	8300
CUST000005	VikramReddy	1	22000
CUST000006	MeeraKrishnan	1	7000
CUST000007	SureshNair	1	6500
CUST000008	KavyaRajan	1	5700
CUST000009	KarthikSubramanian	1	18500
CUST000010	DivyaPillai	1	5200
CUST000011	AdityaVerma	1	7000

CUSTOMER_I	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_AMOUNT
CUST000012	SnehaPatel	1	2450
CUST000014	LakshmiVenkatesan	1	28000
CUST000015	NithinMurugan	1	8350

14 rows selected.